

TFT construction of RCFT correlators (FRS 1)

Outline

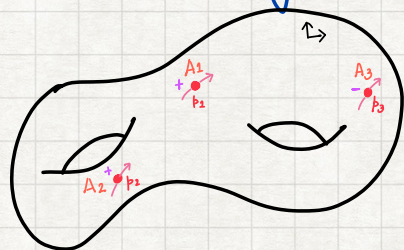
- ↳ 3D TFT
- ↳ Reshetikhin Turaev TFT
- ↳ Websheets $\rightsquigarrow$ obj of $\text{Cob}_{3,2}(\mathcal{C})$
- ↳ FRS construction
- ↳ Example: partition function of torus

NIVEDITA
23 May '22

The category $\text{Cob}_{3,2}(\mathcal{C})$

$\mathcal{C}$ is a modular tensor category

objects: extended surfaces - compact, oriented, closed 2D surface E with the following:



- finite number of marked points

Each marked point has the following structure:

→ a point $p \in E$

→ a germ of arcs $[\gamma]$

such that $\gamma: (-s, s) \rightarrow E$
 $\gamma(0) = p$

→ a choice of an obj in $\mathcal{C}$

→ choice of $\epsilon \in \{-1, 1\}$

- a choice $\lambda \in H_1(E, \mathbb{R})$ where λ is a Lagrangian subspace.

morphisms: $M: E \rightarrow E'$ for given E, E' in $\text{obj Cob}_{3,2}(\mathcal{C})$

- M is compact, oriented 3D manifold such that
 $\partial M \cong -E \sqcup E'$
 with an embedded ribbon graph

A ribbon is an embedding of a rectangle
 $\phi: [0,1] \times [\frac{1}{10}, \frac{9}{10}] \hookrightarrow M$

- core is $[0,1] \times \{0\}$

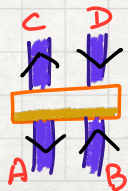
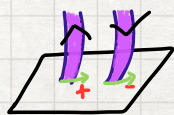
- each ribbon is labelled by an obj of $\mathcal{C}$, ribbon attached to a marked point is assigned the same object of $\mathcal{C}$ as assigned to the marked points.

- outgoing ribbon starts at a $+$ marked point, incoming ribbon ends at $-$ marked point.

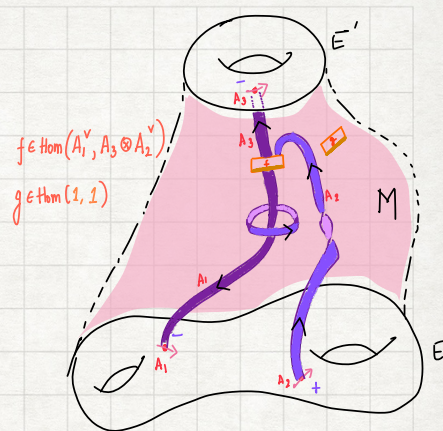
- ribbons can be joined at coupons, an oriented rectangle with two preferred sides - top & bottom
 each coupon is labelled by a morphism from $\mathcal{C}$

[there is a ribbon ending at / starting from each marked point]

- When a ribbon hits a marked point, the edge of the ribbon induces a curve on ∂M passing through the given marked point.
 The germ of this curve should coincide with germ assigned to marked point.



$$f: A^V \otimes B \rightarrow C \otimes D^V$$



3D Topological Field Theory.

A 3D TFT is a projective symmetric monoidal functor from $\text{Cob}_{3,2}(\mathbb{C}) \xrightarrow{Z} \text{Vect}_{\mathbb{C}}$.

i.e.

• every obj in $\text{Cob}_{3,2}(\mathbb{C})$ is assigned a vector space $Z(E)$

• every mor $M: E \rightarrow E'$ is assigned a linear map $Z(M): Z(E) \rightarrow Z(E')$

such that

$$\begin{array}{ccc} \text{[rectangle]} & \xrightarrow{\quad} & \text{id}_{Z(E)} \\ E \times [0, 1] & & \end{array}$$

and,

$$M: E \rightarrow E', \quad f: E' \xrightarrow{C_f} E'', \quad N: E'' \rightarrow E'''$$

(think: twisted cylinder)

$M \sqcup_f N$ is a cobordism between $E \rightarrow E'''$

$$Z(M \sqcup_f N) = \alpha \quad Z(N) \circ Z(C_f) \circ Z(M)$$

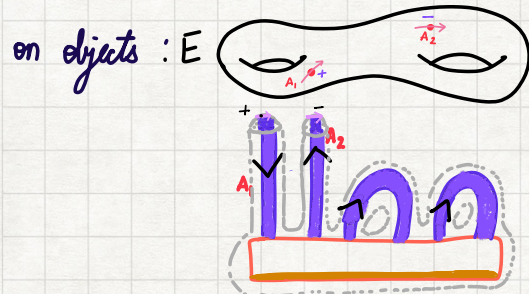
$$\uparrow$$

$$\exists \alpha \in \mathbb{C}^*$$

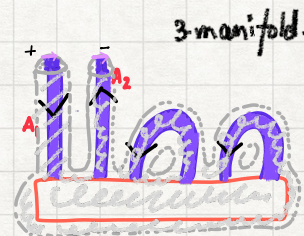
$\uparrow$
associated to the gluing anomaly.

The Reshetikhin-Turaev TFT

$$Z_{RT}^e : \text{Cob}_{3,2}(e) \longrightarrow \text{Vect}_e$$



Given an extended surface, create the following ribbon graph. The boundary of the thickening of this ribbon graph is diffeomorphic to the object we start with.



$$\bigoplus_{i_1, i_2 \in I} ((A_1^V \otimes A_2) \otimes (U_{i_1} \otimes U_{i_1}^V) \otimes (U_{i_2} \otimes U_{i_2}^V)) = \psi(E) \in \text{obj } \mathcal{C}$$

$\psi(E, i_1, i_2)$
 I is the indexing set for simple objects in $\mathcal{C}$. Ways of labelling this graph.

$$Z_{RT}^e(E) = \text{Hom}_{\mathcal{C}}(1, \psi(E)) \text{ is the vector space assigned to object } E.$$

Example: $E = T^2$

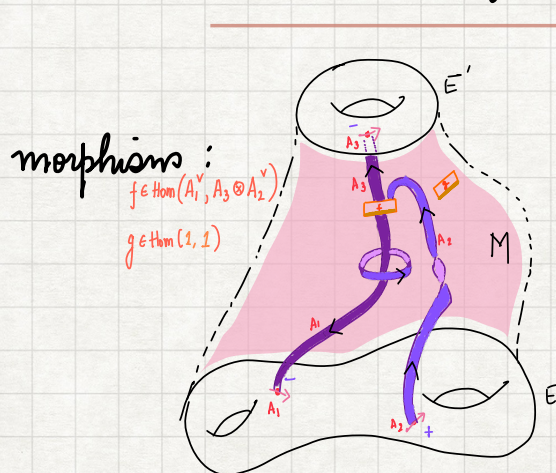
$$\begin{aligned} Z_{RT}^e(T^2) &= \text{Hom}_{\mathcal{C}}(1, \bigoplus_{i \in I} (U_i \otimes U_i^V)) \cong \bigoplus_{i \in I} \text{Hom}_{\mathcal{C}}(1, U_i \otimes U_i^V) \quad (\text{since } I \text{ is finite and using universal property of product}) \\ &\cong \bigoplus_{i \in I} \text{Hom}_{\mathcal{C}}(U_i, U_i) \cong \bigoplus_{i \in I} \mathbb{C} \text{id}_{U_i} = \mathbb{C} \{ \text{id}_{U_i} \mid i \in I \} \text{ free vector space generated by } \text{id}_{U_i} \text{ for } U_i \text{ simple} \end{aligned}$$

$$Z_{RT}^e(T^2) \text{ is generated by } Z_{RT}^e(\text{cobordism } \phi \rightarrow T^2)(1) = |X_i\rangle$$

Let E be a connected extended surface of genus g .
 Define $\eta_E : Z_{RT}^e(E) \rightarrow Z_{RT}^e(E)$ given by $\eta_E|_{\text{hom}_e(1, \psi(E, i_1, \dots, i_g))}$ homogeneous component

$$= \mathbb{Q}^{1-g} \prod_{i=1}^g \dim(U_{i_n}) \text{id}_{\text{hom}_e(1, \psi(E, i_1, \dots, i_g))} \left. \vphantom{\prod_{i=1}^g} \right\} \text{just multiplication by scalar.}$$

$\mathbb{Q}$ is $\sqrt{\sum_{i \in I} \dim(U_i)^2}$. If $E = E_1 \sqcup \dots \sqcup E_k$, $\eta_E = \eta_{E_1} \otimes \eta_{E_2} \cdots \otimes \eta_{E_k}$.



Pick homogeneous elements of $Z_{RT}^e(E)$ and $Z_{RT}^e(E')^V$

From this we obtain 3-manifolds (with embedded ribbon graphs) whose boundary are E and E' .

We interpret these as cobordisms

$$E^{\text{filled}} : \phi \rightarrow E$$

$$E'^{\text{filled}} : E' \rightarrow \phi$$

$$\text{and consider the 3-manifold (closed)} E^{\text{filled}} \sqcup M \sqcup E'^{\text{filled}} = \tilde{M}$$

This has a ribbon graph embedded in it.

We now associate RT-3 manifold invariant to $\tilde{M}$.
 (this is a complex number)

This defines a map $Z_{RT}^e(E) \times Z_{RT}^e(E')^V \rightarrow \mathbb{C}$

and thus we obtain a map $\tilde{Z}_{RT}(M) : Z_{RT}^e(E) \rightarrow Z_{RT}^e(E')$.

$$Z_{RT}^e(M) = \eta_{E'} \circ \tilde{Z}_{RT}(M)$$

The above construction defines a 3D TFT.

Note: RT invariants assigned to closed 3-manifolds can be found in Reshetikhin Turaev (91)

Worldsheets.

We have a special, symmetric Frobenius algebra A in $\mathcal{C}$.

Defn. A worldsheet X is a compact 2D manifold with boundary. It has following data

(i) an orientation

(ii) a set of marked points

* bulk insertion

→ A point of X , $p \in X \setminus \partial X$

→ A germ of arc through p

→ two simple objects U_i, U_j

→ an A - A bimodule homomorphism

$$\phi: U_i \otimes^+ A \otimes^- U_j \rightarrow A$$

* boundary insertion

→ a point $b \in \partial X$

→ a germ of an arc through b (∃ a representative of it whose image lies completely in ∂X)

→ an object B in $\mathcal{C}$

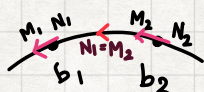
→ two left A -modules M, N

and a left A -module homomorphism

$$\psi: M \otimes B \rightarrow N$$

No two bulk/boundary insertions can be at the same point.

If we have two boundary insertions adjacent to each other $(b_1, [x_1], M_1, N_1, B_1, \psi_1)$ and $(b_2, [x_2], M_2, N_2, B_2, \psi_2)$ then $M_2 = N_1$



(iii) to every connected component of ∂X which has no boundary insertions we assign the left A -module

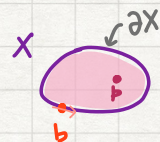
Doubling the Worldsheet

- This process takes a worldsheet X and turns it into $\hat{X}$, an object in $\text{Ob}_{3,2}(\mathcal{C})$

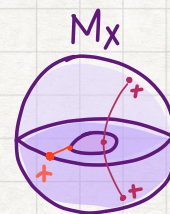
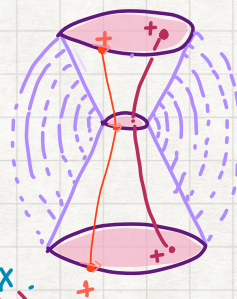
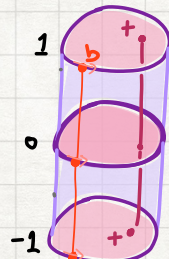
$\hat{X}$ is closed, oriented 2-manifold given by $X \times \{-1, 1\} / \sim$ where $\forall x \in \partial X, (x, 1) \sim (x, -1)$

M_X is a 3-manifold $= X \times [-1, 1] / \sim$, where $\forall x \in \partial X, (x, -1) \sim (x, 1)$
 $\partial M_X = \hat{X}$

object



M_X



$\partial M_X = \hat{X}$

orientation on $\hat{X}$ is that induced from M_X .

bulk field in X $(p, [\gamma], U_i, U_j, \phi)$

- We get 2 marked points on $\hat{X}$
- both are assigned $\varepsilon = +$
- $(p, +1) \rightsquigarrow U_i, (p, -1) \rightsquigarrow U_j$
- one germs are induced similarly

boundary field immersion in X
 $(b, [\gamma], B, M, N, \psi)$

- We get a unique marked point in $\hat{X}$ given by $(b, \pm 1)$, call $\hat{b}$
- The germ of one associated to b gives a germ going through $\hat{X}$
- assigned object B , and $\varepsilon = +$
 $(\hat{b}, [\gamma], B, +)$

* Lagrangian subspace:

$$\hat{X} \xrightarrow{i_2} M_X$$

$$H_1(i_2, \mathbb{R}) : H_1(\hat{X}, \mathbb{R}) \rightarrow H_1(M_X, \mathbb{R})$$

has a bilinear form given by Poincaré duality + $\wedge + \int$ } symplectic structure.
 this vanishes on $\ker H_1(i_2, \mathbb{R})$

Morphism

We want to construct a morphism of $\text{Cob}_{3,2}(e)$ from ϕ to $\hat{X}$.
The underlying 3-manifold of this cobordism is $M_X = X \times [-1, 1] / \sim$

- Consider embedding $L_X : X \longrightarrow M_X$
 $x \longmapsto (x, 0)$

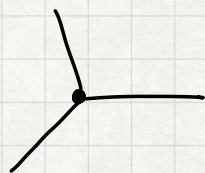
choice #1

A triangulation T of X by polygons of arbitrary many edges, at each vertex only 2/3 edges meet.

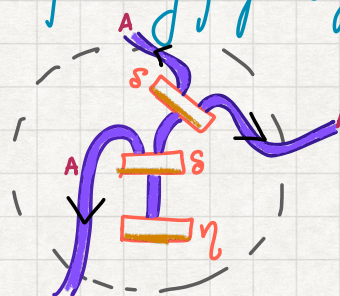
- ∂X is covered by edges of T
- 2-valent vertices of T occur precisely at the marked points
- for a bulk field insertion at p with arc-germ $[X]$, there is a representative of $[X]$ that lies completely in an edge of T .

choice #2

On each 3-valent vertex in the interior, place the following fragment of ribbon graph with 3 outgoing ribbons labelled A.

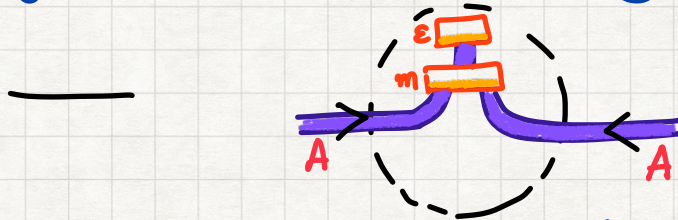


three ways of assignment.



choice #3

for edge which does not lie on the boundary

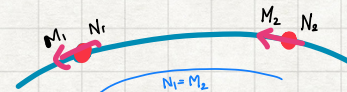


two ways of assignment.

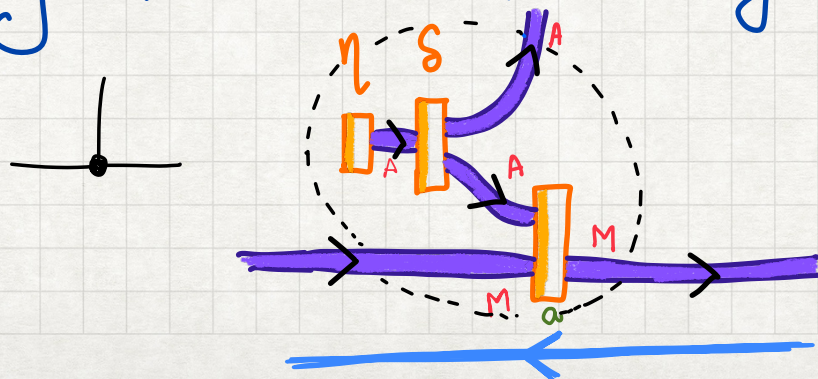
for edges in T which are on the boundary, we assign a left A -module

- if the boundary component in which the edge lies has no boundary field insertions, we assign it the module which was assigned to that boundary.
(arrow given by orientation of the boundary)

- if the boundary component in which the edge lies has boundary field insertion, then the edge necessarily lies between two insertions, so that we assign the left module $N_1 = M_2$



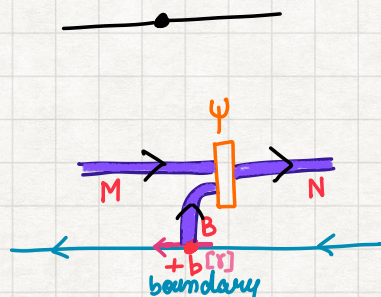
for every 3-valent vertex on the boundary ∂X



orientation of the boundary.

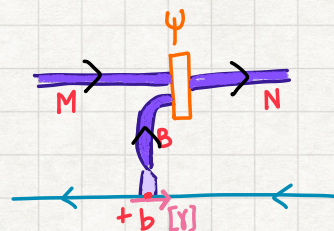
for 2-valent vertex on the boundary
 let the boundary field insertion at the vertex be $(b, [x], M, N, B, \psi)$

Case 1:



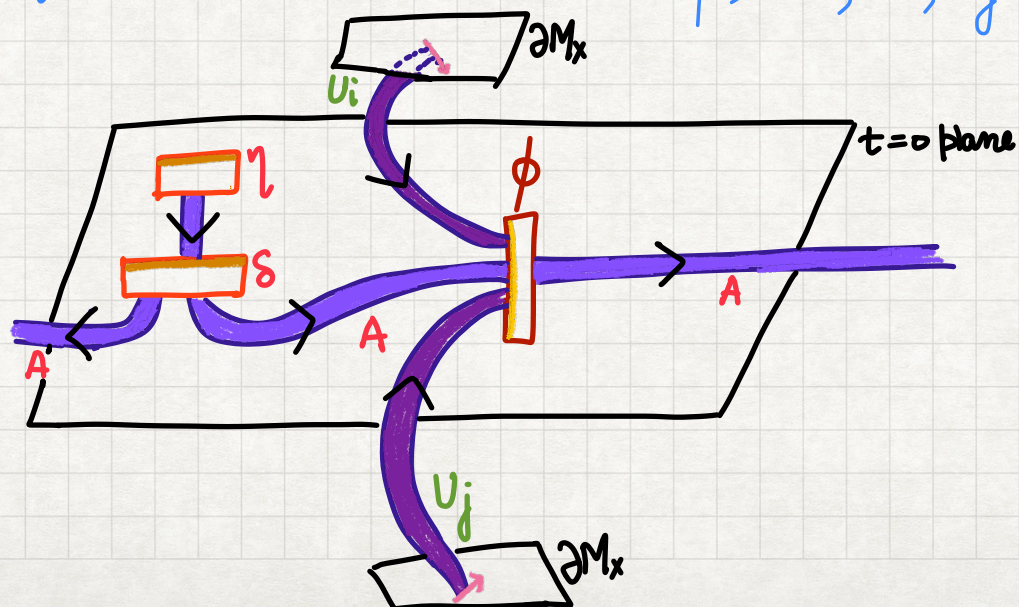
depending on direction of the germ of arc associated to the marked point

Case 2:



There is a marked point on $\hat{x}$
 given by $(\hat{b}, [x], B, +)$
 extending the Bribbon to that point.

2-valent vertex in the interior $(p, [x], v_i, v_j, \phi)$



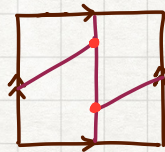
Example : Torus. (worldsheet)

$$X = T^2$$

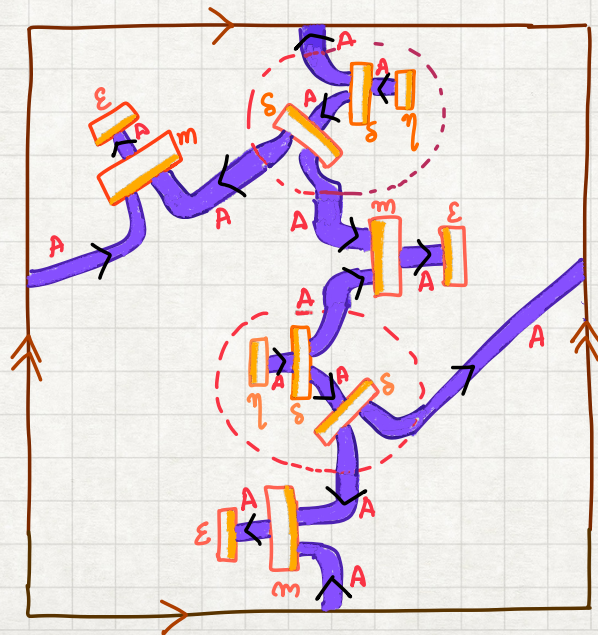
$$\hat{X} = T^2 \times \{-1, 1\} \quad (\text{no quotient as } \partial T^2 = \emptyset) = T^2 \sqcup \overline{T^2}$$

$$M_X = \text{[Diagram of a cylinder with three horizontal sections]} \quad T^2 \times [-1, 1]$$

Choosing a triangulation for T^2



two 3-valent interior intersections
interior edges



Obtaining correlation functions

Given a worldsheet X , we have constructed an object $\hat{X}$ in $\text{Cob}_{3,2}(\mathcal{C})$ and a morphism $M_X : \phi \rightarrow \hat{X}$.

We evaluate $Z_{RT}^{\mathcal{C}}(M_X) : \mathbb{C} \rightarrow Z_{RT}^{\mathcal{C}}(\hat{X})$ and thus obtain an element $Z_{RT}^{\mathcal{C}}(M_X)(1) \in Z_{RT}^{\mathcal{C}}(\hat{X})$.

Conjecture (holography)

Let $\mathcal{V}$ be a rational vertex operator algebra (chiral data). $\text{Rep } \mathcal{V}$ is a modular tensor category. Let $Y \in \text{Obj } \text{Cob}_{3,2}(\text{Rep } \mathcal{V})$.

$$Z_{RT}^{\text{Rep } \mathcal{V}}(Y) \simeq \text{Conformal blocks on } Y!$$

Holomorphic factorization

for simplicity, let $\partial X = \phi$. The $\hat{X} = X \sqcup \bar{X}$.

$$\pi : \hat{X} \rightarrow X$$

We have a map $\text{doub} : M_X \rightarrow M_{\hat{X}}$

$\uparrow$ moduli space of complex structures with marked points.

Thus, an element of $Z_{RT}^{\mathcal{C}}(\hat{X})$ gives an element of conformal blocks on $\hat{X}$. Then using the holomorphic factorization (pull back along doub of conformal block on $\hat{X}$) we get a correlator on X !!

A block on Conformal Blocks.

- A correlator on Y would be a fn $M_Y \rightarrow \mathbb{C}$ where M_Y is the moduli space of complex structures on Y with marked points.
- The action of conformal transformations on the correlator is governed by differential eqns. called Ward Identities. (conservation law)
- The solutions to this are called conformal blocks.

More generally, ??

There is a vector bundle on M_Y (depending on $\mathcal{V}$).

This vector bundle has a connection on it.

- Correlator is a flat section of this vector bundle and conformal blocks are space of all such sections ('pre-correlators').

Torus partition function.

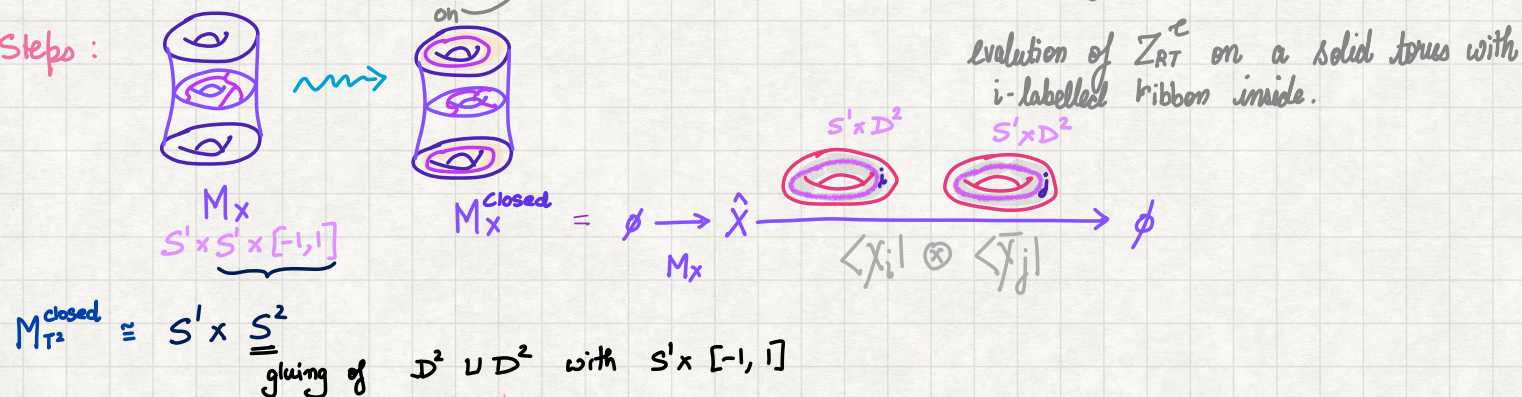
$$X = T^2, \quad \hat{X} = T^2 \sqcup \bar{T}^2$$

$$\begin{aligned} Z_{RT}^e(\hat{X}) &= Z_{RT}^e(T^2) \otimes Z_{RT}^e(\bar{T}^2) = Z_{RT}^e(T^2) \otimes Z_{RT}^e(T^2)^v \\ &= \text{End}_{\text{Vect}_C}(Z_{RT}^e(T^2)) \end{aligned}$$

from RT-TFT construction we have a basis for $Z_{RT}^e(\hat{X})$ given by $|X_i\rangle \otimes |\bar{X}_j\rangle$

$$Z_{RT}^e(M_X)(1) = \sum_{i,j \in I} z_{ij} |X_i\rangle \otimes |\bar{X}_j\rangle.$$

Steps:

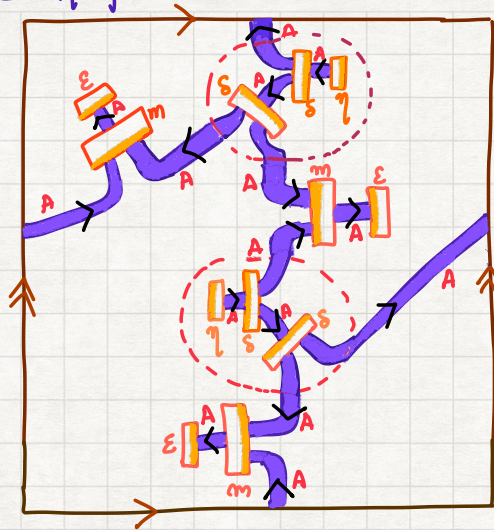


We can simplify the ribbon graph structure in M_X^{closed} using identities and relations on coupons.

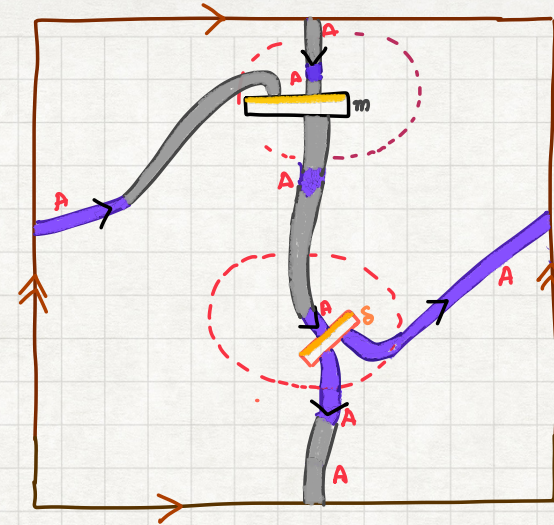
Th: The value of RT-TFT on a manifold does not change under simplifications of the ribbon graph (in our case using properties of A and $MT C$)

↑
Frobenius obj.

Simplifications



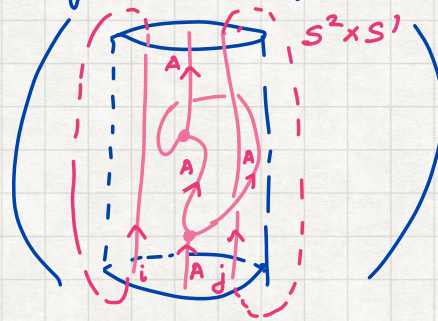
using Frobenius laws we make these simplifications



Similar in structure to the triangulation we started with (which was the "simplest" triangulation).

Similarly, we can simplify the ribbon graph inside M_x^{closed} and obtain

$$z_{ij} = \sum_{\text{RT}}^e$$



Fact: $z_{ij} \in \mathbb{Z}_{\geq 0}$

? To obtain the correlator, we need to map $|X_i\rangle$ to conformal blocks on the torus (no marked points).

$$\begin{aligned} \chi_i : M_{T^2} &\rightarrow \mathbb{C} \\ \tau &\mapsto \chi_i(\tau) \end{aligned} \quad \begin{array}{l} \text{[physicists call this character]} \\ \text{depends on } \mathcal{V} \text{ and its representation } i! \end{array}$$